PARTICLE IMPLEMENTATION

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SNORT IDS / IPS LAB REPORT

PRACTICAL IMPLEMENTATION

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Objective

The goal of this lab was to:

- Deploy Snort as an IDS to detect malicious activity.
- Upgrade the setup to Snort Inline IPS to actively block attacks.
- Validate detection and prevention using real attack simulations from a separate attacker VM.

Environment Setup

- Tools & Platform
- Ubuntu Linux (Snort VM)
- VirtualBox
- Snort 2.9.x
- hping3, telnet, netcat
- Two Virtual Machines:
- Snort VM (IDS/IPS)
- Attacker VM

Lab Environment & Architecture

```
Internet
|
[ enp0s3 ]  --->  Snort (Inline IPS)  --->  [ enp0s8 ]
                                            |
                                            Internal Network
```

virtual Machines:

"Snort VM"
Adapter 1: NAT (enp0s3)
Adapter 2: Internal Network (enp0s8)
Attacker VM
Adapter 1: Internal Network

1.1 Snort Installation & Verification

```
sudo apt update && sudo apt upgrade -y
sudo apt install snort -y
```

1.2 Verify installation:

```
snort -V
```

2.1 Snort Configuration:

```
sudo nano /etc/snort/snort.conf
```

2.2. Network variables:

```
ipvar HOME_NET 10.0.2.0/24
ipvar EXTERNAL_NET any
```

2.3 Enabled local rules:

```
include $RULE_PATH/local.rules
```

IDS Mode Lab

IDS Rules Implementation

Rule 1 – ICMP (Ping Detection)

```
alert icmp any any -> $HOME_NET any \
(msg:"IDS ICMP Ping Detected"; sid:1000001; rev:1;)
```

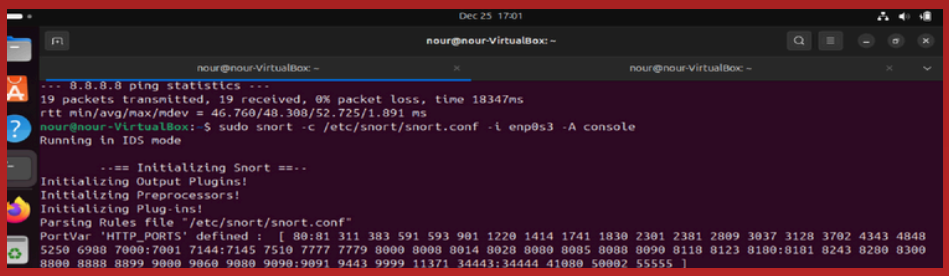
Conf

Rule 2 – HTTP Scan Detection

```
alert tcp any any -> $HOME_NET 80 \
(msg:"IDS HTTP Scan Detected"; sid:1000002; rev:1;)
```

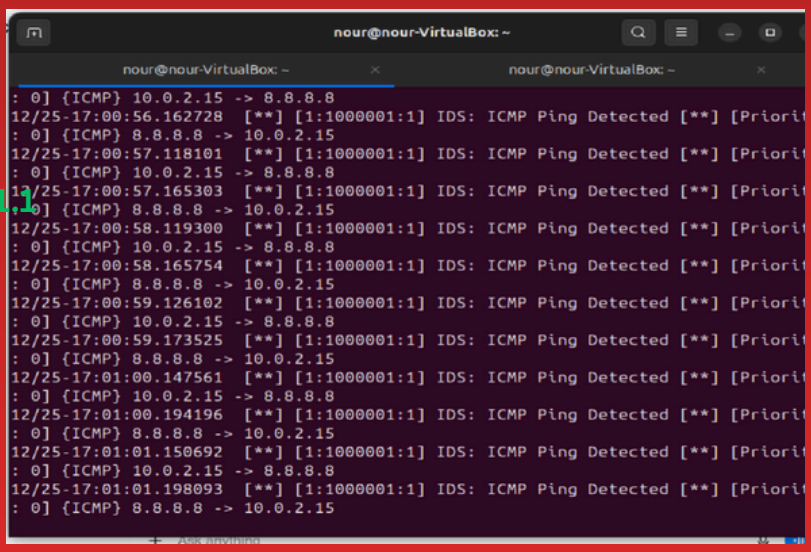
Running Snort as IDS

```
sudo snort -c /etc/snort/snort.conf -i enp0s3 -A console
```



IDS Testing

ping 8.8.8.8
curl http://1.1.1.1



IPS Configuration (Inline Mode)

Enable IP Forwarding

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Enable Inline Policy

In snort.conf:

```
config policy_mode:inline
```

IPS Rule (Drop Rule)

```
sudo snort -c /etc/snort/snort.conf -Q \
--daq afpacket -i enp0s3:enp0s8 -A console
```

Attacker Simulation – IP Configuration

Why IP Configuration Is Required

In this lab, manual IP configuration was required to simulate a real attacker–victim scenario within an isolated internal network.
Since the IPS is running inline, both the attacker and the protected host must be able to communicate through the Snort machine.

Snort VM – Internal Interface (enp0s8)

```
sudo ip addr add 192.168.100.1/24 dev enp0s8
sudo ip link set enp0s8 up
```

Why we did this:

- Assigns a static internal IP address to the Snort IPS interface.
- Allows Snort to act as a gateway and inspection point between the attacker and the internal network.
- Ensures that all traffic targeting the protected host passes through Snort.
- Required for Inline IPS mode to properly forward or drop packets.
- In this lab, 192.168.100.1 represents the protected system behind the IPS.

Attacker VM – Internal Network Interface

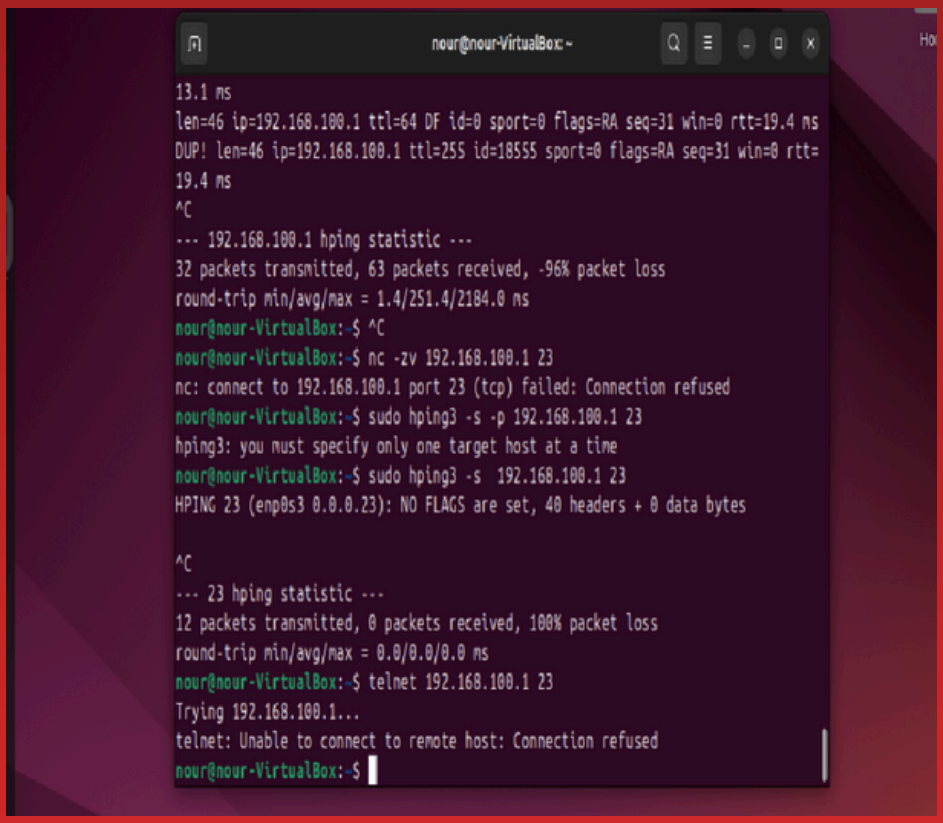
```
sudo ip addr add 192.168.100.2/24 dev enp0s8
sudo ip link set enp0s8 up
```

Why we did this:

- Assigns an IP address in the same internal subnet as the Snort VM.
- Enables direct communication with the IPS-protected host.
- Simulates a real attacker attempting lateral movement or service exploitation.
- Allows generation of malicious traffic (Telnet scans and SYN packets).
- In this lab, 192.168.100.2 represents the attacker machine.

Attack Commands

```
hping3 -S -p 23 192.168.100.1
telnet 192.168.100.1 23
nc -zv 192.168.100.1 23
```



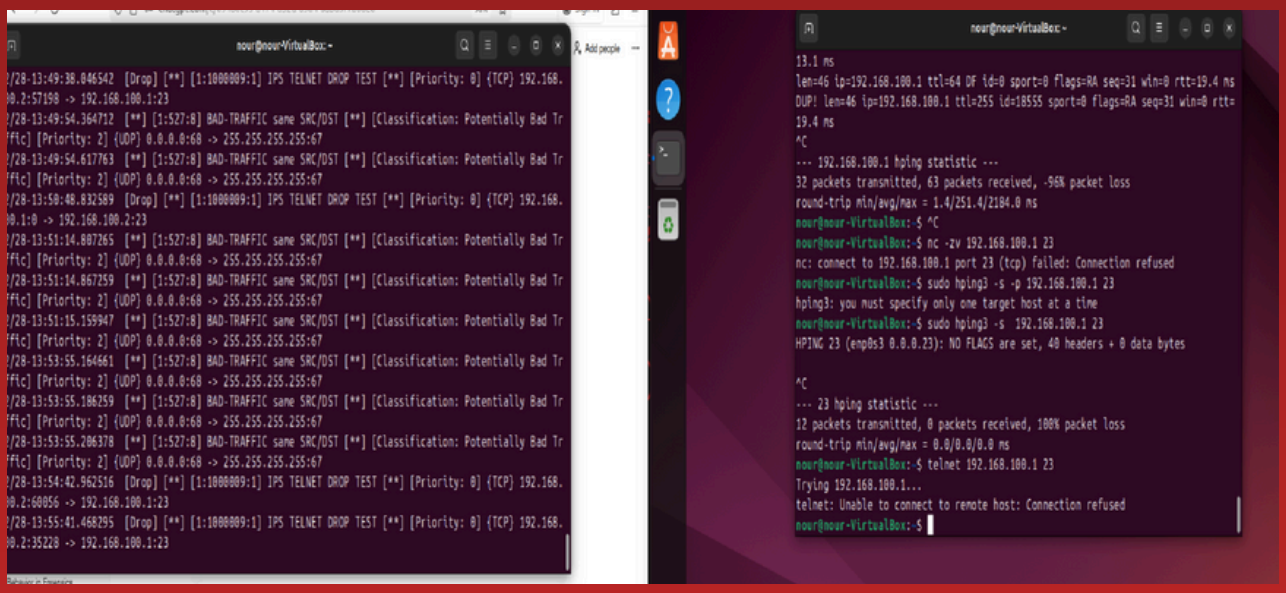
Results & Evidence

Snort logs confirmed successful blocking of the attack attempts while operating in Inline

IPS mode

[Drop] [1:1000009:1] IPS TELNET DROP TEST
{TCP} 192.168.100.2 -> 192.168.100.1:23

the attacker was unable to establish a Telnet connection, while Snort actively dropped the malicious packets in real time.



Snort alert log showing DROP action

Conclusion

This lab successfully demonstrated:

- Snort running as IDS
- Custom detection rule creation
- Transition to Inline IPS
- Active traffic blocking using drop rules
- Real attack simulation and prevention

Snort effectively prevented Telnet access, proving its capability as a functional Inline Intrusion Prevention System (IPS).